

PanCaKES: Panatopic Adversarial Nuclei Classification and Knowledge-Enhanced Segmentation using Pathopix-GANs

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1 Problem and Data Set Description

In the rapidly evolving field of medical image analysis, histopathological imaging stands at the forefront, offering unparalleled insights into disease diagnosis and treatment planning [1] [2]. The PanNuke dataset, a comprehensive collection of annotated histopathological images, serves as a pivotal resource in this domain [3]. Comprising 8,000 images with approximately 100,000 annotated nuclei across various tissue types, it provides a rich foundation for developing advanced image processing techniques [3].

However, the dataset presents challenges such as class imbalance and inherent noise in nuclear structures, which can hinder the performance of traditional deep learning models. To address these issues, we propose a multi-step pipeline that integrates advanced data augmentation, generative modeling, and state-of-the-art segmentation techniques. Our goal is to enhance the dataset’s quality and diversity, thereby improving segmentation accuracy and contributing to more robust medical imaging applications [3] [4].

2 Preliminary Ideas on How to Address the Problem

To overcome the limitations of the PanNuke dataset, we propose the following approach:

2.1 Data Preprocessing and Feature Engineering

We will implement a Generative Adversarial Network (GAN) based approach to synthesize high quality histopathological images [5] [1]. Specifically, a custom PathopixGAN model will be trained to generate realistic nuclear images, focusing on underrepresented cell types. This technique aims to expand the dataset effectively, addressing class imbalance and improving model generalization [1].

In addition, we will employ Stable Diffusion-based models to refine image clarity and reduce noise. These models, originally designed for high-fidelity text-to-image synthesis, will be adapted to preserve structural details in histopathological images while enhancing texture and contrast [6].

2.2 Model Selection and Training

For segmentation, we will leverage the Segment Anything Model v2 (SAM v2) [7], a cutting-edge zero-shot segmentation tool that has demonstrated remarkable accuracy across multiple domains. This model enables precise nuclear segmentation without requiring extensive manual annotation. The refined dataset, enhanced with synthetic augmentation and noise reduction, will serve as an optimal input for training deep segmentation models [7], [8].

We will benchmark segmentation performance using baseline models such as U-Net before transitioning to the enhanced dataset. Performance will be evaluated through metrics like the dice score, f-1, UoI and weighted accuracy, ensuring a comprehensive assessment of model improvements [4].

3 Tools

We plan to use Python (3.12 onwards) for coding, utilizing libraries PyTorch, ONNX, TensorFlow/Keras, OpenCV, and HuggingFace Datasets to facilitate model training and evaluation. We will be using Kaggle GPUs (2*T4 Tesla and 4*L4 Tesla) and Kaggle Notebooks.

4 Expected Results and Evaluation

We hypothesize that integrating GAN-based augmentation, diffusion-based noise reduction, and SAM v2 segmentation will significantly improve segmentation accuracy. We anticipate an increase in the Dice Score and classification accuracy due to the enhanced dataset.

To evaluate our approach, we will use key performance metrics such as:

- **Dice Score:** Measures segmentation accuracy.
- **Precision & Recall:** Assesses nuclear classification reliability.
- **F1 Score:** Provides a balanced evaluation of the model's performance.

Visual assessment of segmentation masks will further validate the qualitative improvements achieved through our methodology.

5 Preliminary Results and Data Set Explored

We have examined the PanNuke dataset, identifying key challenges related to class imbalance in **dead** category and image noise, suppressed class of dead

cell is 1% in the label data. Initial analysis suggests that augmentation and enhancement techniques can significantly improve model performance [4] [9].

6 Outline of Work to Do

1. Implement data preprocessing techniques, including augmentation and noise reduction.
2. Train and validate GAN models for synthetic image generation.
3. Apply Stable Diffusion to enhance dataset quality.
4. Train segmentation models on both raw and enhanced datasets.
5. Evaluate performance using standard medical imaging metrics.
6. Compare results with existing state-of-the-art approaches.

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